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Abel Jansma

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I use machine learning, mathematics, and high-performance computing to understand emergent structures in complex systems with natural and artificial intelligence. I bring experience in interdisciplinary teams at the interface of theory, experiment, and computation.

Experience

Research Fellow | University of Amsterdam | NL **Nov 2024 - Present**
- Received a fellowship from the Dutch Institute of Emergent Phenomena to study the foundations and applications of emergence and higher-order structure in complex systems.
- Affiliated to the Institute of Physics (IoP) and the Institute for Logic, Language and Computation (ILLC).
- Established scientific partnerships with Iliad & Bicameral Labs, US-based AI research institutions.

Research Fellow | Max Planck Institute for Mathematics in the Sciences | GER **Sep 2023 - Sep 2024**
- In the groups of Jürgen Jost and Bernd Sturmfels.

Postdoc | University of Edinburgh | UK **Oct 2022 - July 2023**
- Combined generative models, statistical physics, and causal inference to construct hypergraphs of genetic interactions and discovered novel and rare cell identities in populations of up to 100k transcriptomes.
- Inventor and lead R&D of the Stator software package, in an interdisciplinary team across physics, biology, and informatics, combining fundamental research with biomedical applications.

Information Officer | SciPost | NL **Oct 2017 - July 2018**
- Expanded the editorial board, profiling and contacting potential new editors.

Junior Editor | SPUI25 | NL **Feb 2017 - July 2017**
- Co-organised and presented monthly academic & cultural events for a broad audience.

Distillation Engineer | Mediamatic | NL **Sep 2016 - Feb 2017**
- Designed, built, and demonstrated a bespoke 30 litre vacuum still for artistic and olfactory research.

Education

PhD in Biomedical AI | University of Edinburgh | UK **Sep 2018 - Dec 2022**
- Thesis: *Higher-order interactions in single-cell gene expression: Towards a cybergenetic semantics of cell state*
- Supervised by C. Ponting (Genetics and Cancer), L. Del Debbio (Physics), and A. Khamseh (Informatics).
- Graduated from the *Academy for PhD Training in Statistics* at the universities of Cambridge and Oxford.

MSc Theoretical Physics | University of Amsterdam | NL **Sep 2015 - July 2018**
- Thesis: E_8 *symmetry structures in the Ising model* (supervised by B. Nienhuis)
- Studied nonequilibrium physics and machine learning at the Niels Bohr Institute in Copenhagen.

BSc Physics and Astronomy | University of Amsterdam | NL **Sep 2012 - July 2015**
- Graduated with Honours/Cum Laude and a minor in Computational Science.

Propedeuse in Art and Technology | HKU University of the Arts | NL **Sep 2011 - July 2012**
- Art installations on collective behaviour, exhibited in the Netherlands, Germany, and Finland.

Teaching

Teaching
- Group Theory — MSc Theoretical Physics, 2025, UvA (student evaluation score: 4.8/5)
- Theory of Complex Systems — MSc Theoretical Physics & Computational Science, 2025, UvA (4.1/5)

Supervision

- BSc thesis: *A Layerwise Stability Prior for Fully Online Continual Learning* (Mehmet Polat, 2026, AUC)

Selected Publications & Talks

Articles

- A Compositional Calculus for Semantic Synergy in Language Model Embeddings - Jansma (ICML 2026, workshop on Compositional Learning: Safety, Interpretability, and Agents)
- Decomposing Interventional Causality into Synergistic, Redundant, and Unique Components - Jansma (NeurIPS spotlight, 2025)
- Engineering Emergence - Jansma and Hoel (2025)
- Möbius transforms and Shapley values for vector-valued functions on weighted directed acyclic multigraphs - Forré and Jansma (2025)
- The Fast Möbius Transform: An algebraic approach to information decomposition - Jansma, Mediano, Rosas (Phys. Rev. Res., 2025)
- A Mereological Approach to Higher-Order Structure in Complex Systems: from Macro to Micro with Möbius - Jansma (Phys. Rev. Res., 2025)
- High Order Expression Dependencies Finely Resolve Cryptic States and Subtypes in Single Cell Data - Jansma et al. (Mol. Syst. Biol., 2025)
- Superdense Coding and Stabiliser Codes with Ising-coupled Entanglement - Jansma (preprint, 2024)
- Higher-Order Interactions and Their Duals Reveal Synergy and Logical Dependence beyond Shannon Information - Jansma (Entropy, 2023)
- A Compositional Game to Fairly Divide Homogeneous Cake - Jansma, A (preprint, 2023)

Conferences

- A Causal Calculus of Parts and Wholes - ILIAD conference, USA, 2025 (talk)
- AI Analogy Machines: workshop with Melanie Mitchell, Han v.d. Maas, Jules Hedges, Martha Lewis - NL, 2025 (organiser)
- Higher-order in-and-outeractions - DEMICS23, GER, 2023 (talk)
- A compositional game to fairly divide homogeneous cake - Applied Category Theory, USA, 2023 (poster)
- Model-free estimation of higher-order interactions - CSHL Biology of Genomes conference, USA, 2021 (poster)
- Higher-order interactions in single-cell expression data - European Mathematical Genetics Meeting, FR, 2021 (talk)
- Higher-order interactions in single-cell expression data - CSHL Network Biology conference, USA, 2021 (poster)

Invited Talks

- A Causal Calculus of Parts and Wholes - UvA Computational Science Lab Seminar, NL, 2026
- A Causal Calculus of Parts and Wholes - Softmax Research Gathering, San Francisco, US, 2026
- Solomonoff Induction without Kolmogorov Complexity - The London Initiative for Safe AI, UK, 2026
- From Macro to Micro with Möbius - UvA Institute for Advanced Studies, NL, 2026
- A Causal Calculus of Parts and Wholes - UvA Institute for Advanced Studies, NL, 2025
- A Calculus of Parts and Wholes - DIEP Seminar, NL, 2025
- Quantumness: Physics at different scales - Gramounce Food & Art MA, NL, 2025
- A Compositional approach to higher-order structure in complex systems - Imperial College London, UK, 2024
- Plant Communication: Fact vs. Lore - Multispecies Research Unit, Trust Berlin, GER, 2024
- A unified approach to higher-order structure in complex systems - University of Leipzig, GER, 2024
- From Genetic Hypergraphs to Persistent Quantum Entanglement - Max Planck Institute for Mathematics in the Sciences, GER, 2023
- The information theory of higher-order interactions - UvA Institute for Advanced Studies, NL, 2023
- Cybergenetic networks in the mouse brain: higher-order gene regulation and Boolean logic - IGC Biomedical Genomics section meeting, Dec 2021.
- Towards differential regulation in single-cell expression data - The Liver Growth and Cancer Lab, University of Edinburgh, UK, 2021
- Searching for strange loops in mouse brains - Math. Quantum Physics Seminar, University of Innsbruck, AT, 2021
- Higher-order information lattices - Math. Quantum Physics Seminar, University of Innsbruck, AT, 2021

Open-source Software Development

- PyMergence (2025, 10.5281/zenodo.17210078): a Python package for causal emergence analyses, available at https://github.com/EI-research-group/pymergence

- Stator (2023, 10.1038/s44320-024-00074-1): a Nextflow pipeline for the analysis of higher-order interactions in single-cell data, available at https://github.com/AJnsm/Stator

Grants & Awards

Iliad Research Grant ($125k) | Principles of Intelligence | US — **June 2026 - June 2027**
AI Residency Grant ($8k) | PIBBSS × Iliad | UK — **Jan 2026 - Feb 2026**
Protocol Fellowship Grant ($7.5k) | Ethereum Foundation | GER — **Nov 2022 - March 2023**
Technology Scholarship ($10k) | ASML | NL — **Sep 2015 - Sep 2017**

Volunteering

Co-host | Computational Biology Journal Club — **Feb 2019 - March 2020**

Beekeeper | Anna's Tuin en Ruigte — **Oct 2017 - July 2018**

Member of the Junior Editorial Board | SPUI25 — **Jan - Sep 2017**

Reader | VoorleesExpress — **Nov 2016 - Nov 2017**

- I read books to children to stimulate language development and promote reading.

Committee member | BetaBreak — **Dec 2014 - Jan 2016**

- Moderated and organised public debates on science & society at Amsterdam Science Park.

Languages

Natural	Dutch (native), English (fluent), German (fluent), French (basic)
Programming	Python (SciPy, PyTorch, Qiskit, *etc.*), R, Nextflow, SGE, Git, Haskell, Arduino (C++)